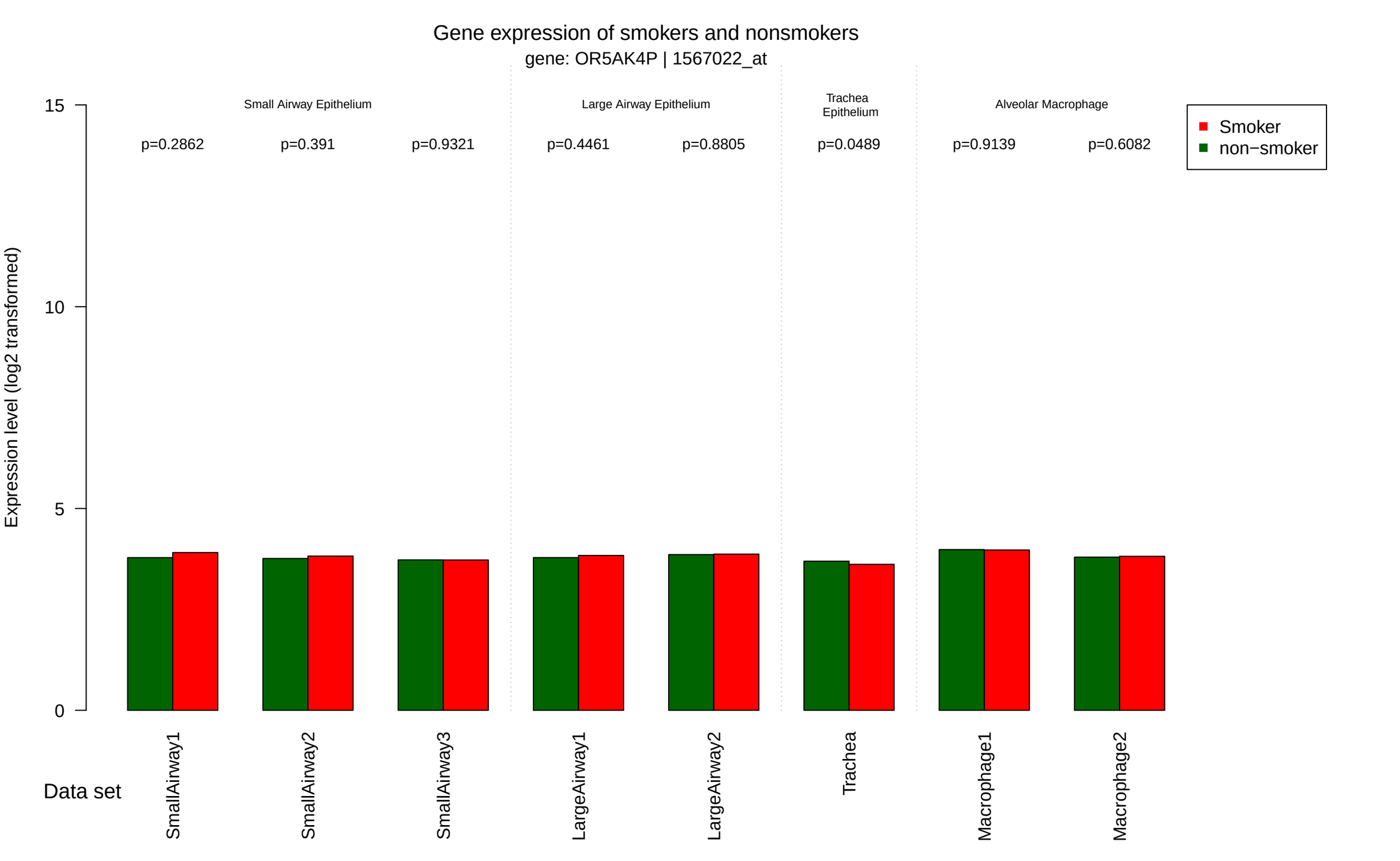
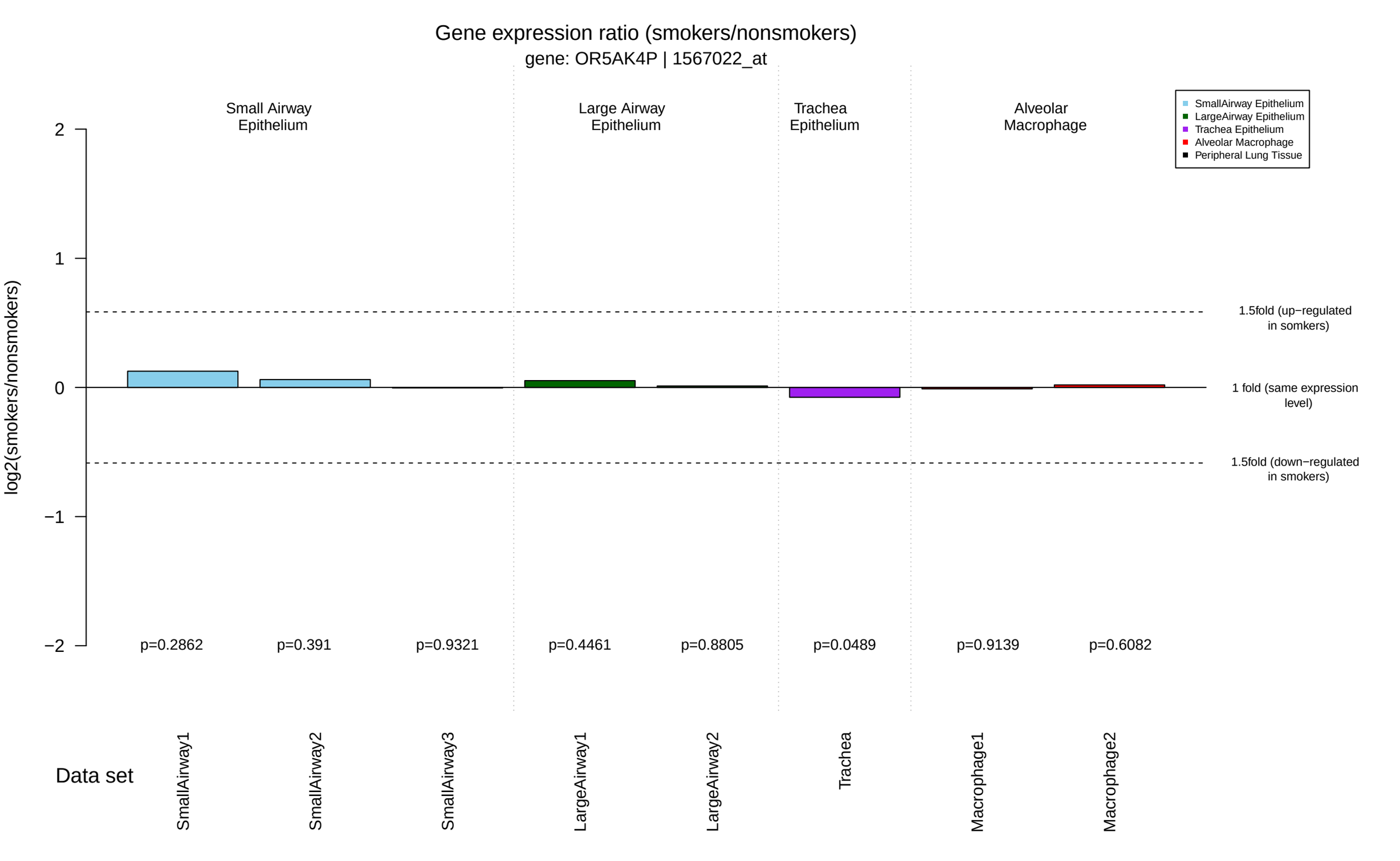
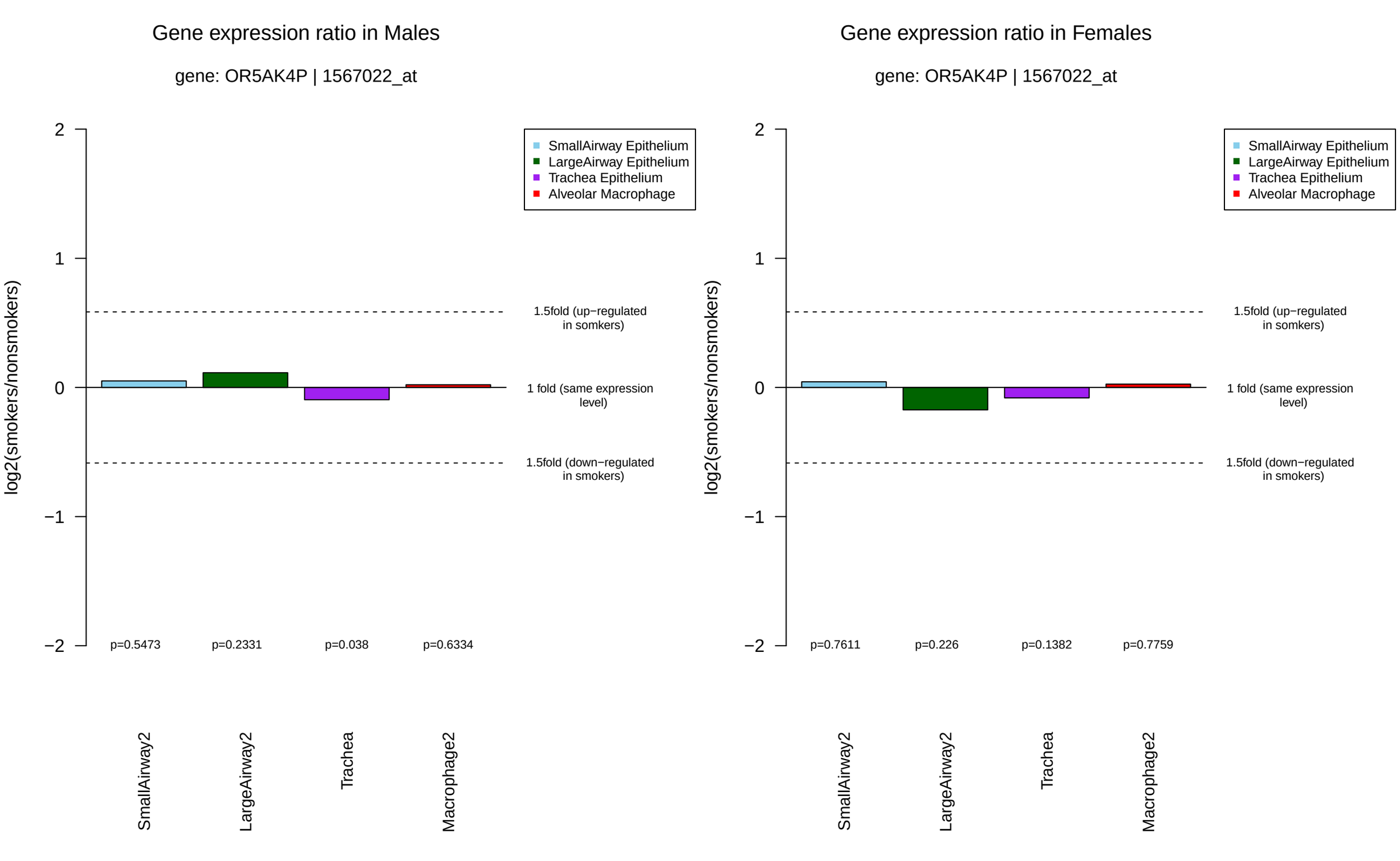
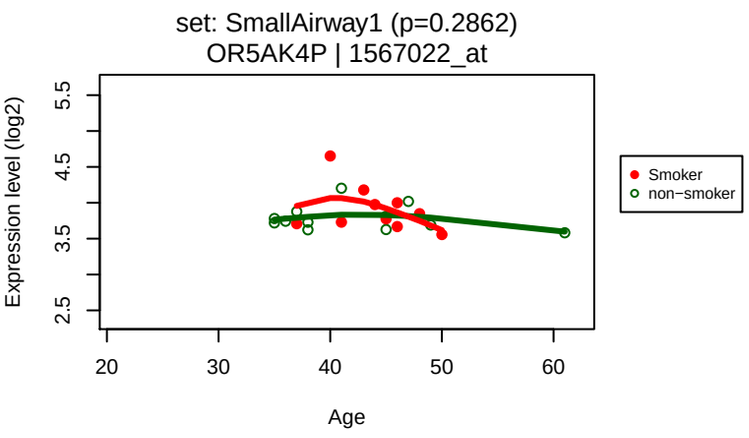
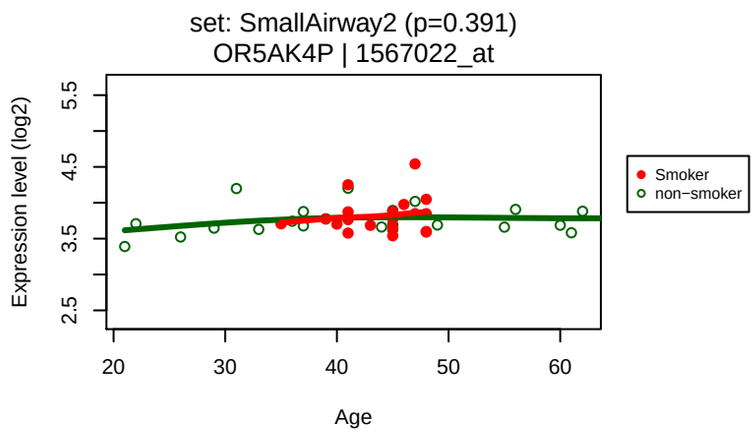
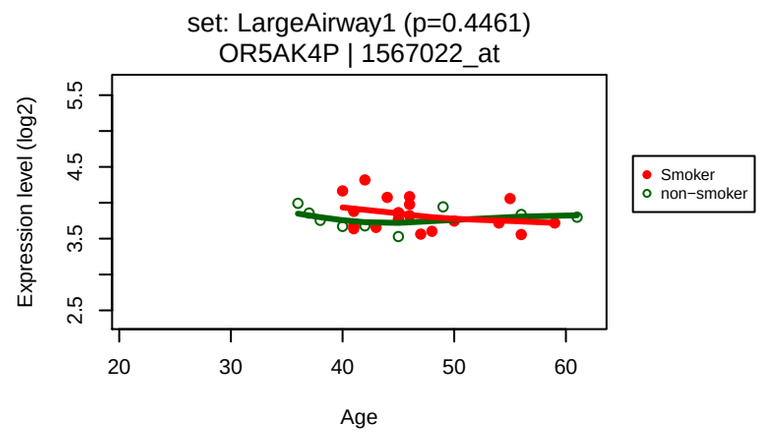

Gene expression of smokers and nonsmokers
gene: OR5AK4P | 1567022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2862
p=0.391
p=0.9321
p=0.4461
p=0.8805
p=0.0489
p=0.9139
p=0.6082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

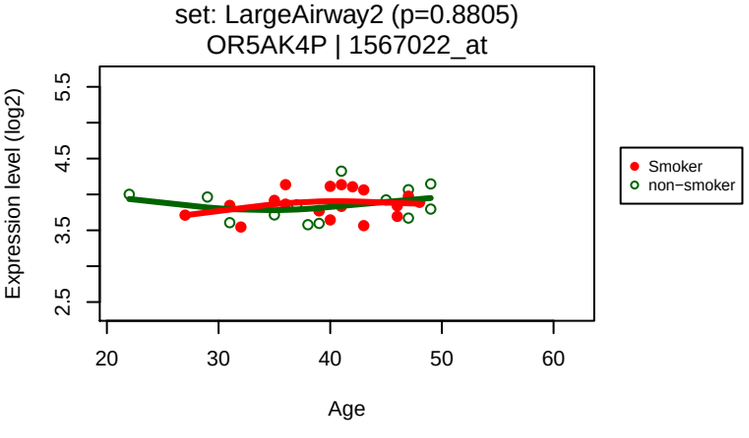

Gene expression ratio (smokers/nonsmokers)
gene: OR5AK4P | 1567022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2862
p=0.391
p=0.9321
p=0.4461
p=0.8805
p=0.0489
p=0.9139
p=0.6082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

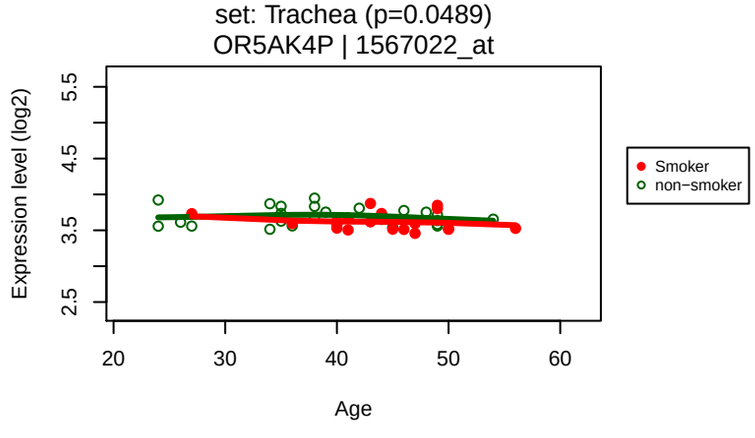

Gene expression ratio in Males
gene: OR5AK4P | 1567022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5473
p=0.2331
p=0.038
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR5AK4P | 1567022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7611
p=0.226
p=0.1382
p=0.7759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

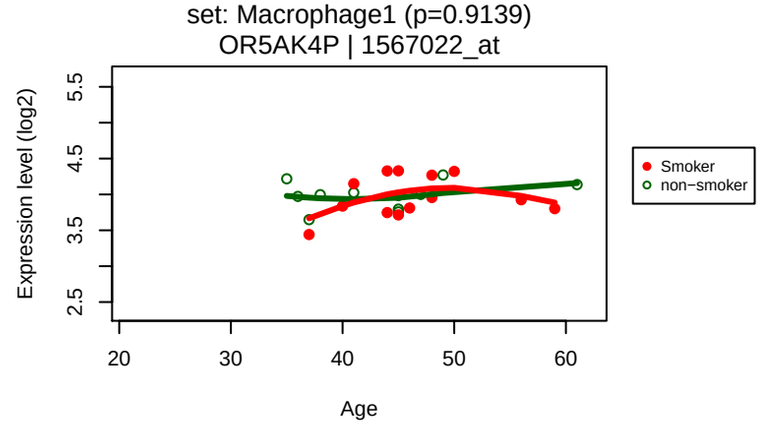

set: SmallAirway1 (p=0.2862)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker

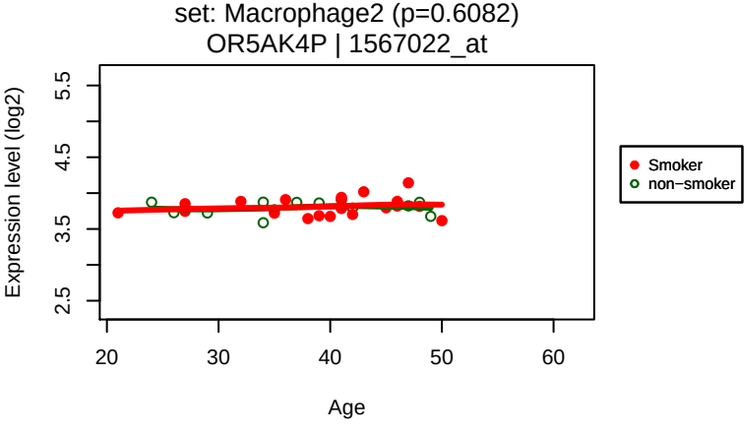

set: SmallAirway2 (p=0.391)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker

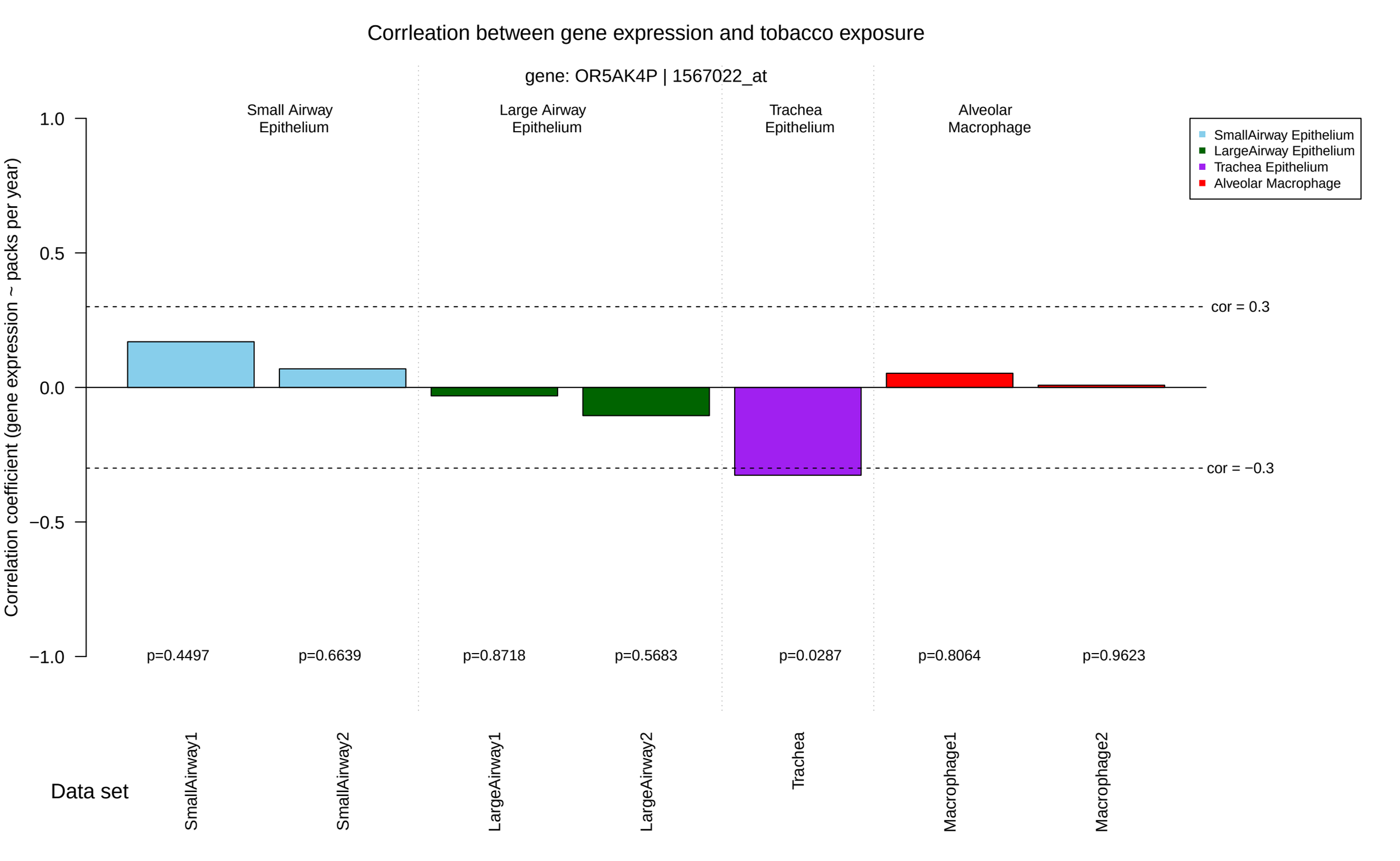

set: LargeAirway1 (p=0.4461)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8805)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0489)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9139)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6082)
OR5AK4P | 1567022_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR5AK4P | 1567022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4497
p=0.6639
p=0.8718
p=0.5683
p=0.0287
p=0.8064
p=0.9623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2